

An explicit odd-order p -group with as many automorphisms as elements

A computer-assisted solution to Kourovka Notebook problem 16.63

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Abstract

We construct a finite group G for which

$$p = 1009, \quad |G| = |\operatorname{Aut}(G)| = 1009^{52359}.$$

The group is the Baker–Campbell–Hausdorff group of an explicitly specified rank-31 Lie ring modulo 1009^{1689} . Its nilpotency class is at most $846 < 1009$. An asymmetric lattice forces every automorphism of the finite Lie ring into an integral logarithm–exponential neighborhood. This gives a bijection between the *full* automorphism set and the kernel of a linear derivation matrix. Exact independently implemented Python and C++ computations establish the Smith valuations of that matrix. The resulting computer-assisted proof answers the existence question affirmatively. The construction and all exact certificates are supplied with separately implemented verification software.

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AI attribution. ChatGPT-6 Astra Pro solved the problem and developed the construction, proof, and supporting scripts, as attributed by the author. This edition was independently checked by Codex using separate AI review agents and exact computational reproduction. These checks are not external human peer review or a proof-assistant formalization.

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1 Statement, sources, and scope

Problem 16.63, attributed to D. MacHale in the September 2026 version of the *Kourovka Notebook*, asks for a nontrivial finite p -group of odd order with $|\operatorname{Aut}(G)| = |G|$ [1]. It appears on printed page 102 and cross-refers to Archive 12.77 on printed page 255. The latter is the separate, now-disproved divisibility conjecture. The notebook website identifies its 1 September 2026 file as the current update. No published resolution of the exact equality question was located in the literature searches recorded in the accompanying ledger.

González-Sánchez and Jaikin-Zapirain construct, for every prime, families with $|\operatorname{Aut}(G)| = O(|G|^{40/41})$ [2]. This explains the negative answer to the archival divisibility problem; it does not itself give the equality required here. Shabani-Attar’s paper treats particular classes: its Proposition 2.5, Theorem 2.6 and Theorem 2.8 exclude odd-prime examples of maximal class, with cyclic Frattini subgroup, and of class two with cyclic centre, respectively [3]. None is a general nonexistence theorem.

The Lie-algebra template used below is the rank-31 example in Omirov–Ruan, Theorem 4.8, with six radical summands [4]. Their arXiv record has version 1, dated 6 May 2026. We specify an integral normalization and establish the required finite-characteristic facts directly; we do not assume that an unspecified characteristic-zero automorphism theorem reduces unchanged modulo p . The asymmetric lattice and finite full-automorphism count are developed in this report.

As elementary controls, a nontrivial abelian group of odd order has the inversion automorphism of order two, so cannot answer the question. For

$$H_p = \langle a, b \mid a^{p^2} = b^p = 1, bab^{-1} = a^{1+p} \rangle,$$

the normal forms $a^u b^v$ give $|H_p| = p^3$. All automorphisms have images $a \mapsto a^u b^v$, $b \mapsto a^{pw} b$, where u is a unit modulo p^2 and $v, w \in \mathbb{F}_p$. Thus $|\operatorname{Aut}(H_p)| = (p-1)p^3$, not p^3 . The exhaustive generating-pair controls give $(27, 54)$, $(125, 500)$ and $(343, 2058)$ for $(|H_p|, |\operatorname{Aut}(H_p)|)$ at $p = 3, 5, 7$. This also checks why counting only a Sylow subgroup would give a false positive.

Theorem 1.1. *There exists a finite group G of order 1009^{52359} whose full automorphism group has order 1009^{52359} . A portable description is given in Definitions 2.1 and 5.1 and equation (15).*

The prime, order and number of generators are not claimed to be minimal. The theorem provides one example, not an assertion for every odd prime.

2 The integral Lie algebra

Fix $p = 1009$. This number is prime, as is also checked by trial division in the independent verifier. All lattices in the proof are over $\mathbb{Z}_p$. For $m \geq 0$, write V_m for the free module of homogeneous binary forms of degree m , with basis $v_i = X^{m-i}Y^i$, $0 \leq i \leq m$. The standard $\mathfrak{sl}_2$ action is

$$[h, v_i] = (m - 2i)v_i, \quad [e, v_i] = iv_{i-1}, \quad [f, v_i] = (m - i)v_{i+1}, \quad (1)$$

with out-of-range terms zero, and $[h, e] = 2e$, $[h, f] = -2f$, $[e, f] = h$.

For binary forms $F \in V_m$, $H \in V_n$, define the *raw transvectant*

$$P_r(F, H) = \sum_{k=0}^r (-1)^k \binom{r}{k} \frac{\partial^r F}{\partial X^{r-k} \partial Y^k} \frac{\partial^r H}{\partial X^k \partial Y^{r-k}}. \quad (2)$$

No factorial normalization is suppressed. On basis vectors the coefficient of the vector indexed by $i + j - r$ in V_{m+n-2r} is

$$\sum_{k=0}^r (-1)^k \binom{r}{k} (m-i)_{r-k} (i)_k (n-j)_k (j)_{r-k}, \quad (3)$$

where $(a)_b = a(a-1)\cdots(a-b+1)$, $(a)_0 = 1$, and $(a)_b = 0$ when $b > a \geq 0$.

Definition 2.1. Let B have ordered basis

$$h, e, f, \quad U_0, \dots, U_6, \quad V_0, \dots, V_4, \quad W_0, \dots, W_6, \quad T_0, T_1, T_2, \quad R_0, \dots, R_4, \quad Z_0.$$

Its six radical summands U, V, W, T, R, Z are copies of $V_6, V_4, V_6, V_2, V_4, V_0$, respectively. In addition to (1), the only nonzero radical brackets, up to skew symmetry, are

Inputs	Output summand	Map
U, U	W	P_3
U, V	T	P_4
U, W	T	P_5
U, R	T	P_4
V, V	T	P_3
V, R	Z	P_4
R, R	T	P_3

The target of a transvectant is identified with the indicated copy of its binary-form module. All other brackets are zero.

This is a complete specification with integer coefficients. In particular, $\text{rank } B = 3 + 7 + 5 + 7 + 3 + 5 + 1 = 31$.

Lemma 2.2. *The displayed bracket is a Lie bracket over $\mathbb{Z}$. Over $\mathbb{Z}_p$, $\mathbb{Q}_p$ and $\mathbb{F}_p$ it is perfect and has centre precisely the Z summand. If $N = U \oplus V \oplus W \oplus T \oplus R \oplus Z$, then*

$$[N, N] = W \oplus T \oplus Z, \quad [N, [N, N]] = T, \quad [N, T] = 0.$$

Proof. The action and transvectants are $\mathfrak{sl}_2$ -equivariant. Hence the only potentially nonzero radical Jacobi expression is the alternating expression on three U inputs, taking values in T . Over the rationals,

$$\Lambda^3 V_6 \cong V_{12} \oplus V_8 \oplus V_6 \oplus V_4 \oplus V_0,$$

as follows by multiplying the seven weight characters and taking the third exterior power. It has no V_2 summand. The equivariant Jacobi map to $T = V_2$ is therefore zero. Its coefficients are integers, so this proves the integer identity. Independently, both implementations check all $\binom{31}{3} = 4495$ basis triples, with integer arithmetic.

The nontrivial simple summands have no $\mathfrak{sl}_2$ -fixed vector at $p = 1009$. Thus the centre is contained in Z , which is central. Every nontrivial summand is generated by its brackets with $\mathfrak{sl}_2$, and $[V, R] = Z$, so B is perfect. The stated lower central terms of N follow directly from the table. Surjectivity here can be checked on highest weight vectors; coefficients such as 720, 8640 and 576 are units at 1009. $\square$

2.1 Two generators of the ambient lattice

Set

$$x = h, \quad y = e + f + U_1 + V_0 + R_2, \quad t = U_0. \quad (4)$$

Lemma 2.3. *The Lie subalgebra over $\mathbb{Z}_p$ generated by x, y is B .*

Proof. The four h -weights in y are $2, -2, 4, 0$. Interpolation in $\text{ad } h$ isolates $e, f, U_1 + V_0, R_2$, with unit denominators at p . The Casimir operator $h^2 + 2h + 4fe$ on modules has eigenvalues 48 on U and 24 on V ; it separates U_1 and V_0 . Applying e, f generates all of U, V, R , using only the nonzero integers at most six as divisors. Their brackets then generate W, T, Z by Lemma 2.2. The modular Lie-closure computations independently obtain dimension 31. Equivalently, finitely many Lie words have a coefficient determinant that is a unit in $\mathbb{Z}_p$, and hence form a lattice basis of B . $\square$

3 The full finite-field automorphism group

In this section put $k = \mathbb{F}_p$, and use the same letters for reductions to k . The assertions about module intertwiners follow directly from (1): the distinct weights lie between -6 and 6 , and the raising/lowering coefficients are units. The indicated modules are simple, nonisomorphic for different highest weights, and have scalar endomorphisms.

Lemma 3.1. *For $m = 0, 2, 4, 6$, one has $H^1(\mathfrak{sl}_2(k), V_m) = 0$.*

Proof. For a cocycle write $\alpha = c(h)$, $\beta = c(e)$, $\gamma = c(f)$. The cocycle equations are

$$(h - 2)\beta = e\alpha, \quad (h + 2)\gamma = f\alpha, \quad \alpha = e\gamma - f\beta.$$

Subtract a coboundary to remove the nonzero-weight components of α . For $m = 2r > 0$, the remaining component is av_r . The weight-two component of the first equation forces $ar = 0$, so $a = 0$. Then $\beta = bv_{r-1}$ and $\gamma = cv_{r+1}$. The last equation gives $(r + 1)(c - b) = 0$, so $c = b$. This is the coboundary of $(b/r)v_r$. All divisions are by units at 1009. For $m = 0$, the equations immediately force $\alpha = \beta = \gamma = 0$.

The independent checks give cocycle/coboundary dimension pairs

$$(0, 0), \quad (3, 3), \quad (5, 5), \quad (7, 7)$$

for these four highest weights. $\square$

Every automorphism of $\mathfrak{sl}_2(k)$ is induced by conjugation by a matrix in $\text{PGL}_2(k)$. For completeness, preservation of the Killing form sends a nonzero nilpotent matrix to a nonzero nilpotent matrix. Conjugate to fix e . The relations then force

$$h \mapsto h + ae, \quad f \mapsto f - \frac{a}{2}h - \frac{a^2}{4}e,$$

which is another matrix conjugation. On V_m , for even m , the compatible PGL_2 action is $\text{Sym}^m(k^2) \otimes \det^{-m/2}$. The determinant weights in (2) make this an action on B . Denote it by ρ .

Let τ fix h, e, f, U, W, T , swap V and R by corresponding basis vectors, and send Z_0 to $-Z_0$. The table shows that τ is an automorphism.

Proposition 3.2. *Every automorphism of $B \otimes k$ has the form*

$$\exp(\operatorname{ad} n) \rho(s) \tau^\epsilon, \quad n \in N, \quad s \in \operatorname{PGL}_2(k), \quad \epsilon \in \{0, 1\}. \quad (5)$$

Here the exponential is a polynomial of degree at most three. In particular, this describes the full automorphism group, including its finite components.

Proof. N is the solvable radical: any solvable ideal maps to a solvable ideal in the simple quotient $\mathfrak{sl}_2(k)$ and so lies in N . After removing the quotient automorphism using ρ , the image of $\mathfrak{sl}_2$ is a complement to N inducing the identity on the quotient. Its graph modulo $N' = [N, N]$ is a one-cocycle into $N/N' \cong V_6 \oplus V_4 \oplus V_4$. Lemma 3.1 allows conjugation by $\exp(\operatorname{ad} n_1)$ to remove this graph. The remaining graph lies in the abelian ideal $N' \cong V_6 \oplus V_2 \oplus V_0$ and is again a one-cocycle, removable by $\exp(\operatorname{ad} n_2)$. Since N has class three, the product of these inner exponentials is another $\exp(\operatorname{ad} n)$; all denominators are units. This is an explicit complement argument, not an appeal to a positive-characteristic Levi theorem without hypotheses.

It remains to classify automorphisms ϕ fixing $\mathfrak{sl}_2$ pointwise. They are equivariant. The characteristic submodule N' implies

$$U \mapsto aU + bW, \quad W \mapsto a^2W, \quad T \mapsto cT, \quad Z \mapsto dZ,$$

and the pair (V, R) is mixed by an invertible two-by-two matrix M . The T component in $[\phi U, \phi U]$ is $2abP_5$, so $b = 0$. The bracket $[U, W]$ gives $c = a^3$. The T components of the brackets on $V \oplus R$, and the brackets with U , give, for $w = (1, 1)$,

$$M^T M = a^3 I_2, \quad wM = a^2 w. \quad (6)$$

Since M is invertible, also $MM^T = a^3 I_2$. As $ww^T = 2 \neq 0$, these imply $2a^4 = 2a^3$, hence $a = 1$. An orthogonal matrix fixing w acts by $+1$ or -1 on its one-dimensional orthogonal complement. Thus $M = I$ or the swap matrix. The Z bracket gives $d = \det M$. Consequently $\phi = 1$ or τ . $\square$

Lemma 3.3. *Over k and over $\mathbb{Q}_p$, all derivations of B are inner, and $\dim \operatorname{Der}(B) = 30$.*

Proof. Let D be a derivation and $\pi : B \rightarrow \mathfrak{sl}_2$ the quotient map. The restriction $\pi D|_N$ kills N' and is equivariant, since $[D(s), n] \in N$. There is no equivariant map from $N/N' = V_6 \oplus V_4^2$ to $\mathfrak{sl}_2 = V_2$; hence $D(N) \subseteq N$. Subtract an inner derivation to make the induced derivation on the quotient zero. The restriction to $\mathfrak{sl}_2$ is then a cocycle into N ; Lemma 3.1, also valid over $\mathbb{Q}_p$, removes it by another inner derivation.

For the remaining equivariant derivation write $D(U) = aU + bW$, $D(W) = 2aW$. As before $b = 0$, and $D(T) = 3aT$. Writing M for the action on $V \oplus R$, the linearized bracket equations are

$$M + M^T = 3aI_2, \quad wM = 2aw.$$

Multiplying by w, w^T gives $8a = 6a$, hence $a = 0$. Then M is skew and annihilated by w , so $M = 0$; also $D(Z) = 0$. Thus the remaining derivation vanishes. The centre has dimension one, so the space of inner derivations has dimension $31 - 1 = 30$.

Independently, each implementation finds rank 931 for the 14415 by 961 derivation matrix of B modulo 1009, and rank 30 for its inner subspace. This directly verifies the finite-field conclusion. $\square$

4 A flag with no nonidentity automorphism

Define the flag

$$\mathcal{F}_2 = \langle x, y \rangle_k \subset \mathcal{F}_3 = \langle x, y, t \rangle_k. \quad (7)$$

Proposition 4.1. *The stabilizer of (7) in the full $\text{Aut}(B \otimes k)$ is the identity. Its infinitesimal stabilizer in $\text{Der}(B \otimes k)$ is zero.*

Proof. Use (5). The plane $\mathcal{F}_2$ projects isomorphically to $P = \langle h, e + f \rangle \subset \mathfrak{sl}_2$, while $\mathcal{F}_3 \cap N = \langle U_0 \rangle$. On N/N' , the inner exponential is the identity and τ fixes U . Therefore s preserves the highest-weight line in U , so belongs to the upper Borel of $\text{PGL}_2(k)$. It also preserves P .

The Killing-orthogonal complement of P is $\langle e - f \rangle$. For a representative $\begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix}$, preservation of this line forces $b = 0$ and $a^2 = 1$. Thus $s = 1$ or $r = \text{diag}(-1, 1)$ projectively.

Suppose $s = r$. A flag-preserving automorphism must send h to h and y to $-y$, since the projection of $\mathcal{F}_2$ is an isomorphism. Modulo N' , fixing h forces $\bar{n}$ to have weight zero. On U_i , r acts as $(-1)^{i+3}$, and in particular fixes U_1 . The U component of the equation for y would require

$$(e + f)\bar{n}_U = -2U_1.$$

But $\bar{n}_U$ is a multiple of U_3 , and $(e + f)U_3 = 3U_2 + 3U_4$. This is impossible.

Now suppose $s = 1$ and $\epsilon = 1$. Fixing h, y similarly requires

$$(e + f)\bar{n} = R_0 - V_0 + V_2 - R_2$$

on $V \oplus R$. The left side, applied to a weight-zero vector, has only weights 2, -2; the right side has nonzero weight-four and weight-zero components. This too is impossible. The only remaining possibility is an inner exponential fixing x, y . They generate B , so it is the identity.

For the infinitesimal assertion use Lemma 3.3, writing $D = \text{ad } u$. Its quotient component $s \in \mathfrak{sl}_2$ preserves P and the highest line in U , hence $s \in \langle h, e \rangle$. The conditions $[s, h] \in P$ and $[s, e + f] \in P$ then force $s = 0$. Now $D(\mathcal{F}_2) \subseteq \mathcal{F}_2$ and its image in the quotient is zero. Since $\mathcal{F}_2 \cap N = 0$, $D(x) = D(y) = 0$, and $D = 0$ by generation. $\square$

5 The asymmetric lattice and its finite quotients

Definition 5.1. Use the following ordered $\mathbb{Z}_p$ -basis $b_0, \dots, b_{30}$ of B :

$$\begin{aligned} &x, y, t, f, U_1, U_2, U_3, U_4, U_5, U_6, \\ &V_0, V_1, V_2, V_3, V_4, W_0, W_1, W_2, W_3, W_4, W_5, W_6, \\ &T_0, T_1, T_2, R_0, R_1, R_2, R_3, R_4, Z_0. \end{aligned}$$

It is unimodular relative to the original basis. Put

$$w_0 = w_1 = 0, \quad w_2 = 1, \quad w_j = 2 \quad (3 \leq j \leq 30), \quad A = \bigoplus_{j=0}^{30} \mathbb{Z}_p p^{w_j} b_j, \quad L = p^2 A.$$

In particular $p^2 B \subset A \subset B$. Although A need not itself be a Lie subring, L is one: $[L, L] \subset p^4 B \subset p^2 A = L$. For $i \geq 1$ write $\bar{L}_i = L/p^i L$. The basis $\ell_j = p^{2+w_j} b_j$ identifies its additive group with $(\mathbb{Z}/p^i \mathbb{Z})^{31}$. Every additive endomorphism of this finite Lie ring is automatically $\mathbb{Z}/p^i \mathbb{Z}$ -linear, so the matrix description includes every Lie-ring automorphism.

Let μ be the bracket of B and let $\nu = p^2\mu$ on the module A . Under $a \mapsto p^2a$, (A, ν) is isomorphic to L . Let

$$\mathcal{E} = \text{End}_{\mathbb{Z}_p}(A), \quad \mathcal{E}_0 = \text{End}_{\mathbb{Z}_p}(B), \quad J = \mathcal{E} \cap p\mathcal{E}_0.$$

For a linear map D , write

$$\delta_\mu(D)(u, v) = D\mu(u, v) - \mu(Du, v) - \mu(u, Dv), \quad \delta_\nu = p^2\delta_\mu.$$

Lemma 5.2. *Let $i \geq 7$. Every automorphism of $\bar{L}_i$ has a lift $Q \in 1+J$. Every solution of $\delta_\nu(D) \equiv 0 \pmod{p^i}$ on A has a lift $D \in J$. Here a lift means an arbitrary integral matrix representative in $\text{GL}(A)$ or $\mathcal{E}$, respectively.*

Proof. For an automorphism lift $Q \in \text{GL}(A)$, its bracket error satisfies

$$E_Q(A, A) \subseteq p^{i-2}A, \quad E_Q(u, v) = Q\mu(u, v) - \mu(Qu, Qv).$$

Since $p^2B \subset A$, bilinearity gives

$$E_Q(B, B) \subseteq p^{i-6}A \subseteq p^{i-6}B. \quad (8)$$

This is integral when $i \geq 6$. The images Qx, Qy lie in $A \subset B$. Induction on Lie words using (8) now shows that Q maps B into B . By Lemma 2.3, finitely many such words span B . The determinant of Q is a unit, independently of basis, so $Q(B) = B$. For $i \geq 7$, its reduction is an automorphism of B/pB .

A map preserving both A and B preserves

$$(A + pB)/pB = \mathcal{F}_2, \quad ((p^{-1}A \cap B) + pB)/pB = \mathcal{F}_3.$$

Proposition 4.1 therefore gives $Q \equiv I \pmod{p\mathcal{E}_0}$. Together with $Q - I \in \mathcal{E}$, this is $Q \in 1 + J$.

For a derivation solution, replace E_Q by $\delta_\mu(D)$. The same word induction, now using the Leibniz error, gives $D(B) \subset B$. Its reduction modulo p is a derivation preserving the flag. The infinitesimal part of Proposition 4.1 gives $D \equiv 0 \pmod{p}$ on B , so $D \in J$. $\square$

6 Integral logarithms count the full automorphism set

The next step is essential: merely counting derivations or known inner automorphisms would not determine the full automorphism group.

Lemma 6.1. *Matrix exponential and logarithm give inverse bijections $J \leftrightarrow 1 + J$. For every $i \geq 3$, they respect congruence modulo $p^i\mathcal{E}$ in both directions.*

Proof. The inclusions of lattices imply

$$p^2\mathcal{E}_0 \subseteq \mathcal{E} \subseteq p^{-2}\mathcal{E}_0. \quad (9)$$

For $X \in J$, every X^r preserves A , and also $X^r \in p^r\mathcal{E}_0 \subseteq p^{r-2}\mathcal{E}$. In the exponential and logarithm series, denominators of terms of degree less than p are units. For terms of degree at least $p = 1009$, the bounds

$$v_p(r!) \leq r/(p-1), \quad v_p(r) \leq r/(p-1)$$

show that all terms still lie in $\mathcal{E}$ and tend to zero there. They lie in $p\mathcal{E}_0$ after removing the exponential's constant term. Thus both series preserve the indicated sets. The usual formal inverse

identities hold in the convergent matrix algebra over $\mathbb{Q}_p$. In particular $1 + J$ consists of invertible maps preserving A ; its inverse also follows from the convergent geometric series.

For precision, suppose $X, Y \in J$ and $X - Y \in p^i \mathcal{E}$. Each term in the telescoping expression for $X^r - Y^r$ lies in $p^i \mathcal{E}$. By (9) it also lies in $p^{i+r-3} \mathcal{E}_0 \subseteq p^{i+r-5} \mathcal{E}$. For $r < p$, division by the relevant denominator loses no precision. For $r \geq p$, $r - 5$ exceeds both $v_p(r!)$ and $v_p(r)$. The same conclusions therefore hold after division and summation, for both series. Finally $p^i \mathcal{E} \subset J$ when $i \geq 3$, so the quotient bijections are well defined. $\square$

Put

$$\mathcal{T} = \text{Hom}_{\mathbb{Z}_p}(\Lambda^2 A, A), \quad \mathcal{T}_0 = \text{Hom}_{\mathbb{Z}_p}(\Lambda^2 B, B).$$

The bounds $p^2 B \subset A \subset B$ give

$$p^2 \mathcal{T}_0 \subset \mathcal{T} \subset p^{-4} \mathcal{T}_0, \quad p^6 \text{End}(\mathcal{T}_0) \subset \text{End}(\mathcal{T}) \subset p^{-6} \text{End}(\mathcal{T}_0). \quad (10)$$

Lemma 6.2. *For $D \in J$ and $Q = \exp D$, Q preserves the bracket ν modulo $p^i \mathcal{T}$ if and only if $\delta_\nu(D) \in p^i \mathcal{T}$.*

Proof. Let Q act on bracket tensors by $\rho_{\mathcal{T}}(Q)\eta = Q\eta(Q^{-1}, Q^{-1}\cdot)$. The condition $\rho_{\mathcal{T}}(Q)\nu - \nu \in p^i \mathcal{T}$ is equivalent to the ordinary bracket-preservation condition: precompose both inputs with Q , which is invertible on A and preserves $p^i \mathcal{T}$. Its differential $X = \rho_{\mathcal{T},*}(D)$ is the operator

$$X\eta = D\eta - \eta(D\cdot, \cdot) - \eta(\cdot, D\cdot).$$

Here $X \in \text{End}(\mathcal{T}) \cap p \text{End}(\mathcal{T}_0)$ and $X^r \in p^{r-6} \text{End}(\mathcal{T})$ for $r \geq 6$. Over $\mathbb{Q}_p$, the output and two input actions commute, whence $\rho_{\mathcal{T}}(\exp D) = \exp X$. Factor

$$\exp X - I = XU(X), \quad U(X) = \sum_{r \geq 0} \frac{X^r}{(r+1)!}.$$

For $r < p - 1$, the denominator is a unit. For $r \geq p - 1$, the valuation $r - 6 - v_p((r+1)!)$ is positive. Thus $U(X)$ preserves $\mathcal{T}$. Moreover $X^7 \in p \text{End}(\mathcal{T})$, so modulo p the series for U is a polynomial in a nilpotent operator, with constant term I . Consequently $U(X)$ is invertible on $\mathcal{T}$. It commutes with X . It follows that

$$(\exp X - I)\nu \in p^i \mathcal{T} \iff X\nu = \delta_\nu(D) \in p^i \mathcal{T}.$$

There is no loss of congruence precision in this equivalence. $\square$

Proposition 6.3. *For $i \geq 7$, exponential and logarithm give a bijection of sets*

$$\text{Aut}(\bar{L}_i) \longleftrightarrow \ker(\delta_\nu : \mathcal{E}/p^i \mathcal{E} \longrightarrow \mathcal{T}/p^i \mathcal{T}). \quad (11)$$

Proof. Every automorphism and every derivation solution lies in the respective logarithm-exponential domain by Lemma 5.2. Lemma 6.1 gives a bijection on representatives modulo $p^i \mathcal{E}$, and Lemma 6.2 gives precisely the required equivalence of nonlinear and linear bracket conditions. These statements apply to *all* automorphisms, not just to a subgroup. $\square$

No assertion that the bijection is an isomorphism of additive groups is needed or made.

7 The exact Smith computation

Let $c_{ij,k}^b$ be the structure constants of B in the adapted basis. In the $\ell_j = p^{2+w_j}b_j$ basis of L , the constants are

$$c_{ij,k}^L = p^{2+w_i+w_j-w_k}c_{ij,k}^b. \quad (12)$$

All exponents are between zero and six. These constants, with $i < j$, are also provided as ordinary integers in `data/group_lie_presentation.json`.

The matrix K for δ_ν has rows (i, j, k) with $i < j$, and columns (a, b) for the matrix entry D_{ab} . Explicitly,

$$K_{(i,j,k),(a,b)} = \mathbf{1}_{k=a}c_{ij,b}^L - \mathbf{1}_{i=b}c_{aj,k}^L - \mathbf{1}_{j=b}c_{ia,k}^L. \quad (13)$$

Thus K is an integer matrix of shape 14415×961 .

Proposition 7.1 (Exact computational certificate). *The rank of K over $\mathbb{Q}_p$ is 931. Its nonzero p -primary Smith valuations are*

Valuation	0	1	2	3	4
Multiplicity	87	55	758	6	25

Their sum is 1689; the nullity is 30.

Certificate and verification. The Python implementation constructs the integer brackets from (3), changes basis, builds (13), and performs local Smith elimination modulo p^{12} . Each pivot has minimum p -valuation in the remaining block. Only unit inverses are taken; divisions by a pivot power of p are checked to be exact. The resulting plan consists of 931 original row indices, column indices and valuations.

A separately written C++ implementation independently reconstructs the brackets, the basis transformation and the entire matrix from the formulas. It does not read the Python coefficient tables. It verifies the proposed plan modulo p^6 , checking divisibility of the active block at each new valuation level, pivot normalization, exact elimination, and a zero residual block. It finds the same five multiplicities. The Python implementation also replays the plan at precision six.

Every reported valuation is less than six, so it is determined exactly by this finite-ring calculation. Integer Jacobi gives 30 independent inner derivations in the rational kernel, hence rank at most $961 - 30 = 931$. The 931 nonzero pivots give rank at least 931. There can therefore be no additional nonzero invariants hidden at higher precision. All matrix operations are exact elementary operations over the indicated local ring. The sum is $55 + 2(758) + 3(6) + 4(25) = 1689$. $\square$

Corollary 7.2. *For every $i \geq 7$,*

$$|\mathrm{Aut}(\bar{L}_i)| = p^{30i+1689}. \quad (14)$$

Proof. In Smith coordinates, each of the 30 zero diagonal entries contributes p^i kernel elements; each nonzero entry p^v contributes p^v because $i > 4 \geq v$. Combine Proposition 7.1 with the full-set bijection (11). $\square$

8 The finite group and the equality

Now set $i = 1689$ and define

$$G = \mathrm{BCH}(\bar{L}_{1689}). \quad (15)$$

This is an explicit construction, not a library identifier: its elements are the 31-tuples modulo 1009^{1689} in the ℓ basis, with the Lie bracket (12). Its group product is the BCH Lie polynomial truncated after bracket length 846.

Lemma 8.1. *The Lie ring $\bar{L}_{1689}$ has order p^{52359} and nilpotency class at most 846.*

Proof. L is free of rank 31 over $\mathbb{Z}_p$, so its quotient has precisely $p^{31 \cdot 1689} = p^{52359}$ elements, with unique additive coordinates. Since $L \subset p^2 B$ and B is a Lie lattice, $\gamma_m(L) \subset p^{2m} B$ for every m . On the other hand $p^{1689} L = p^{1691} A$ contains $p^{1693} B$. Consequently

$$\gamma_{847}(L) \subset p^{1694} B \subset p^{1693} B \subset p^{1689} L,$$

which proves the class bound. $\square$

For clarity, BCH here can be defined without any software convention. Take the terms of degree at most 846 in $\log(\exp X \exp Y)$ in the completed free associative algebra over $\mathbb{Q}$ and express each homogeneous term as a Lie polynomial. For example, the right-normed Dynkin operator divided by the degree provides such an expression. Evaluate these Lie polynomials using the displayed bracket. The denominators have only prime factors at most 846, so all coefficients have unique interpretations modulo 1009^{1689} . This is a finite, canonical rule for multiplication; it need not be expanded into a very large word list to specify the group.

The Lazard correspondence applies to finite p -Lie rings of class strictly less than p [5, 6]. It preserves the underlying set, group order, and the full automorphism group; see in particular the discussion on pages 433–434 of [6]. Thus the BCH product is associative, has identity zero and inverse $-u$, and every group automorphism corresponds to a Lie-ring automorphism. No exponent- p hypothesis is required. The actual class bound is $846 < 1009$, so these are the applicable hypotheses, rather than a general assertion beyond the Lazard range.

Proof of Theorem 1.1. Lemma 8.1 proves the exact order and validates the BCH construction. Corollary 7.2 and the full-automorphism preservation give

$$|\mathrm{Aut}(G)| = |\mathrm{Aut}(\bar{L}_{1689})| = 1009^{30 \cdot 1689 + 1689} = 1009^{52359} = |G|.$$

The prime is odd and the group is nontrivial. This answers the stated existence question affirmatively. $\square$

9 Reproducibility and verification boundaries

9.1 Fast independent verification

From the root of the bundle, run

```
python3 scripts/resume.py
```

This compiles and runs the independent C++ verifier and also tests that a corrupted pivot certificate is rejected. Alternatively, with a GCC or Clang compiler supporting 128-bit integers, run

```
mkdir -p build
c++ -std=c++17 -O2 -o build/independent_verify \
    scripts/independent_verify.cpp
./build/independent_verify data/smith_pivot_plan.txt
```

The C++ verification uses only the standard library and the supplied pivot plan. It reconstructs all mathematical input coefficients itself. No GAP, SageMath, Magma, remote computation, or randomized group identification is needed for this check.

9.2 Regeneration

The independent Python construction requires NumPy and SymPy. The exact commands are

```
python3 scripts/lie31.py
python3 scripts/exact_linear.py
python3 scripts/smith_local.py --precision 12
python3 scripts/smith_local.py --verify-plan --precision 6
python3 scripts/export_presentation.py
python3 scripts/sanity_groups.py
```

The bundle includes executed logs, the coefficient tables, both independent implementations, the Smith pivot plan, and the unscaled rank certificate. The original supplied run used Python 3.13.5, NumPy 2.3.5, SymPy 1.14.0, and g++ 14.2.0 (Debian 14.2.0-19). Independent full regeneration for this edition used Python 3.12.14, NumPy 2.3.5, SymPy 1.14.0, and Apple Clang 21.0.0 on macOS 26.6.2 (arm64). The only verifier repair was an indentation-only portability correction for Clang. The strict build used `-std=c++17 -O2 -Wall -Wextra -Werror`. The C++ local modulus was the exact integer $1009^6 = 1055229678769825441$; products use 128-bit arithmetic. The measured times in each log describe its recorded environment only. Fresh records are in `audit/reproduction/`. A third implementation using arbitrary-precision Python integers independently reconstructs the brackets, checks the exported presentation and a 931-square rank minor, and verifies the Smith data at precision five. A separate implementation checks the finite-field generation, involution, cohomology, and flag ranks. The C++ verifier also passes an address- and undefined-behavior-sanitized build. The independent audit reports record the scope of each check.

9.3 What the computations do and do not certify

The finite computations verify Jacobi, generation, cohomology dimensions, derivation ranks, infinitesimal flag ranks, and all local Smith invariants. The full finite-field stabilizer, the lifting lemma, the integral logarithm–exponential correspondence, and the application of Lazard are proved above. The full automorphism count follows from the combination.

The group has not been materialized as a Cayley table, nor has a general purpose package enumerated all its automorphisms. Such enumeration is unnecessary for the exact upper and lower bounds. The BCH polynomial of length 846 has not been expanded into a full monomial list; the construction is specified by the canonical finite polynomial and the explicit Lie table. Independent implementation means separately written code reconstructing the input, not independent human peer review. No formal proof assistant or external reviewer is claimed.

A Selected inspected output

The final independent C++ run includes

```
original_integer_jacobi_checks=4495
adapted_integer_jacobi_checks=4495
unscaled_derivation_rank_mod_p=931 derivation_dimension=30
smith_valuation=0 multiplicity=87
smith_valuation=1 multiplicity=55
smith_valuation=2 multiplicity=758
smith_valuation=3 multiplicity=6
smith_valuation=4 multiplicity=25
sum_smith_valuations=1689
candidate_group_order_exponent=52359
predicted_aut_order_exponent=52359
ALL_CHECKS_PASSED
```

Some long log lines are split here for readability. The names `candidate_group_order_exponent` and `predicted_aut_order_exponent` are retained from the computational checkpoint; the proof establishes the corresponding group statements. A separate negative test changes the first proposed pivot valuation from zero to five. The verifier exits nonzero with a pivot-valuation mismatch.

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